

Linux Filesystem

What is file system?

A file system is the method and data structure that an operating system uses to control how data is stored and retrieved. Without a file system, information placed in a storage medium would be one large body of data with no way to tell where one piece of information stops and the next begins. The Linux File System is a structured method of storing and organizing data on a Linux machine. It arranges files in a hierarchical directory format starting from the root directory (/).

What do you mean by storage configuration in linux?

Storage configuration in Linux is the process of setting up hard drives and partitions so the operating system can save and read files.

Exercise 1 – Identify Your Linux Filesystem

Explore the storage configuration of your Linux system.

Determine:

1. The storage devices available.

```
msis@master:/$ lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
loop0        7:0      0    4K  1 loop /snap/bare/5
loop1        7:1      0   74M  1 loop /snap/core22/2411
loop2        7:2      0   74M  1 loop /snap/core22/2437
loop3        7:3      0  66.8M  1 loop /snap/core24/1587
loop4        7:4      0  11.1M  1 loop /snap/firmware-updater/167
loop5        7:5      0  66.8M  1 loop /snap/core24/1643
loop6        7:6      0 614.5M  1 loop /snap/gnome-46-2404/164
loop7        7:7      0 260.3M  1 loop /snap/firefox/8763
loop8        7:8      0 257.6M  1 loop /snap/firefox/8664
loop9        7:9      0 606.1M  1 loop /snap/gnome-46-2404/153
loop10       7:10     0 531.5M  1 loop /snap/gnome-42-2204/263
loop11       7:11     0  16.5M  1 loop /snap/firmware-updater/226
loop12       7:12     0  11.8M  1 loop /snap/snap-store/1390
loop13       7:13     0  395M  1 loop /snap/mesa-2404/1165
loop14       7:14     0 531.4M  1 loop /snap/gnome-42-2204/247
loop15       7:15     0  402M  1 loop /snap/mesa-2404/1839
loop16       7:16     0  15.7M  1 loop /snap/snap-store/1367
loop17       7:17     0  50.1M  1 loop /snap/snapd/27710
loop18       7:18     0  91.7M  1 loop /snap/gtk-common-themes/1535
loop19       7:19     0  50.1M  1 loop /snap/snapd/27591
loop20       7:20     0   828K  1 loop /snap/snapd-desktop-integration/391
loop21       7:21     0 219.9M  1 loop /snap/thunderbird/1201
loop22       7:22     0   828K  1 loop /snap/snapd-desktop-integration/387
loop23       7:23     0 221.1M  1 loop /snap/thunderbird/1228
nvme0n1      259:0     0 953.9G  0 disk
├─nvme0n1p1  259:1     0   200M  0 part /boot/efi
├─nvme0n1p2  259:2     0    16M  0 part
├─nvme0n1p3  259:3     0 299.2G  0 part
├─nvme0n1p4  259:4     0   811M  0 part
├─nvme0n1p5  259:5     0   400G  0 part
├─nvme0n1p6  259:6     0    16G  0 part
└─nvme0n1p7  259:7     0 237.7G  0 part /
```

This command shows disks, partitions, sizes and mount point in a tree structure

Disk: The physical storage drive hardware (like an SSD or HDD) connected to your computer.

Partition: A logical section or isolated slice of a disk used to separate data.

Size: The total data capacity allocated to a physical disk or logical partition.

Mount Point: The directory folder where a partition is attached to be viewed and accessed.

Real-world use: An administrator uses lsblk to identify whether a server has disks such as /dev/sda, /dev/nvme0n1 , USB drives or additional storage

2. The partitions available on each device.

```
nsis@master:/$ lsblk -f
NAME FSTYPE FSVER LABEL      UUID                                FSAVAIL FSUSE% MOUNTPOINTS
loop0
  squash 4.0                                0    100% /snap/bare/5
loop1
  squash 4.0                                0    100% /snap/core22/2411
loop2
  squash 4.0                                0    100% /snap/core22/2437
loop3
  squash 4.0                                0    100% /snap/core24/1587
loop4
  squash 4.0                                0    100% /snap/firmware-updater/167
loop5
  squash 4.0                                0    100% /snap/core24/1643
loop6
  squash 4.0                                0    100% /snap/gnome-46-2404/164
loop7
  squash 4.0                                0    100% /snap/firefox/8763

nvme0n1
├─nvme0n1p1
│   vfat  FAT32              78F5-57AC              119.4M   39% /boot/efi
├─nvme0n1p2
├─nvme0n1p3
│   ntfs              16BAF5BDBAF59A07
├─nvme0n1p4
│   ntfs              C6C21AEEC21AE28B
├─nvme0n1p5
│   ntfs              New Volume 1C285026285000E6
├─nvme0n1p6
│   swap             1                  d9ed9cd0-32b1-431a-a906-36a23630ddf7
└─nvme0n1p7
   ext4             1.0              e436143f-1e5f-4ede-babc-19beb509ae52 157.9G   27% /
```

Adds filesystem type, UUID, and mount point.

Real-world use: Useful when troubleshooting boot problems or determining which filesystem is mounted on a particular partition.

3. The filesystem type used by each partition.

```
msis@master:/$ df -T
Filesystem      Type      1K-blocks    Used Available Use% Mounted on
tmpfs           tmpfs      1603000       3076   1599924    1% /run
/dev/nvme0n1p7  ext4     244179276 66106384 165596428   29% /
tmpfs           tmpfs      8014996         4    8014992    1% /dev/shm
tmpfs           tmpfs       5120          8        5112    1% /run/lock
efivarfs        efivarfs    246          178         64   74% /sys/firmware/efi/efivars
tmpfs           tmpfs      8014996         0    8014996    0% /run/qemu
/dev/nvme0n1p1  vfat     200704       78476   122228    40% /boot/efi
tmpfs           tmpfs     1602996       2556   1600440    1% /run/user/1000
```

Displays filesystem type, total size, used space, available space, and mount point.

Real-world use: A security analyst checks whether a filesystem is nearly full because malware logs, large dumps, or suspicious files may consume disk space.

4. The device/partition on which `/` is located.

```
msis@master:/$ df -Th /
Filesystem      Type  Size  Used Avail Use% Mounted on
/dev/nvme0n1p7  ext4  233G   64G  158G   29% /
```

This directly tells you the filesystem containing `/`.

Real-world use:

Useful when determining which disk/partition contains the operating system.

5. The filesystem type used for `/`.

```
msis@master:/$ findmnt /
TARGET
SOURCE      FSTYPE OPTIONS
/ /dev/nvme0n1p7 ext4    rw,relatime
```

6. The total, used, and available storage space.

Device/Partition	Filesystem type	Size	Mount Point
nvme0n1p1	vfat	200M	/boot/efi
nvme0n1p2	-	16M	-
nvme0n1p3	ntfs	299.2G	-
nvme0n1p4	ntfs	811M	-
nvme0n1p5	ntfs	400G	-
nvme0n1p6	swap	16G	-
nvme0n1p7	ext4	237.7G	/

Exercise 2 – Disk → Partition → Filesystem → Mount Point

Explain in 2–3 sentences the relationship between these four components.

Relationship

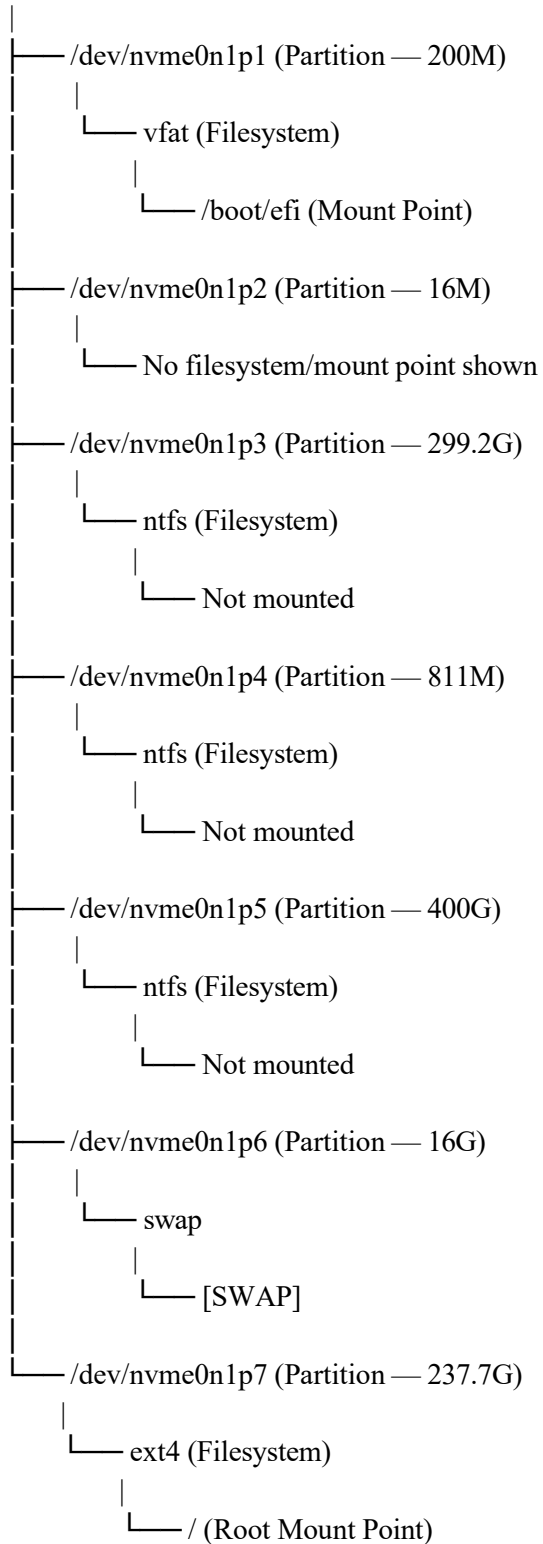
A **disk** is the physical or virtual storage device. A **partition** is a defined section of that disk, a **filesystem** organizes data within the partition, and a **mount point** is the location in the Linux directory tree where that filesystem becomes accessible.

Real-world cybersecurity use:

During forensic investigation, identifying this chain helps an investigator determine exactly where evidence is physically/virtually stored.

Using the information obtained in Exercise 1, draw the storage organization of your own Linux system.

dev/nvme0n1 (Disk — 953.9G)



Exercise 3 – Explore the Root Filesystem

Start from your current location.

1. Determine your current working directory.

```
msis@master:~$ pwd  
/home/msis
```

pwd-print working directory

2. Navigate to the root directory.

```
msis@master:~/Day1$ cd /  
msis@master:/$
```

3. Display the contents of the root directory.

```
msis@master:/$ ls  
bin          dev          lib64        mnt          root         snap         tmp  
bin.usr-is-merged  etc          lib.usr-is-merged  nfs_share    run          srv          usr  
boot         home         lost+found   opt          sbin         swap.img    var  
cdrom        lib          media        proc         sbin.usr-is-merged  sys
```

4. Identify all directories directly available under `/`.

```
msis@master:/$ ls -ld /  
drwxr-xr-x 24 root root 4096 Aug 11 13:50 /
```

5. Determine whether hidden entries are present.

```
msis@master:/$ ls -la /
total 4194408
drwxr-xr-x 24 root root      4096 Aug 11 13:50 .
drwxr-xr-x 24 root root      4096 Aug 11 13:50 ..
lrwxrwxrwx  1 root root         7 Apr 22  2024 bin -> usr/bin
drwxr-xr-x  2 root root      4096 Feb 26  2024 bin.usr-is-merged
drwxr-xr-x  4 root root      4096 Aug 25 09:58 boot
dr-xr-xr-x  2 root root      4096 Aug  5  2025 cdrom
drwxr-xr-x 20 root root      4960 Sep  1 17:44 dev
drwxr-xr-x 155 root root     12288 Sep  1 18:03 etc
drwxr-xr-x  8 root root      4096 Sep  1 10:07 home
lrwxrwxrwx  1 root root         7 Apr 22  2024 lib -> usr/lib
lrwxrwxrwx  1 root root         9 Apr 22  2024 lib64 -> usr/lib64
drwxr-xr-x  2 root root      4096 Apr  8  2024 lib.usr-is-merged
drwx----- 2 root root     16384 Apr 21 20:26 lost+found
drwxr-xr-x  4 root root      4096 Apr 24 16:19 media
drwxr-xr-x  3 root root      4096 Aug 25 16:33 mnt
drwxrwxrwx  2 root root      4096 Aug 11 13:50 nfs_share
drwxr-xr-x  4 root root      4096 Aug  7 10:52 opt
dr-xr-xr-x 536 root root         0 Sep  1 17:44 proc
drwx-----  7 root root      4096 Sep  1 16:08 root
drwxr-xr-x 45 root root      1280 Sep  1 18:00 run
lrwxrwxrwx  1 root root         8 Apr 22  2024/sbin -> usr/sbin
drwxr-xr-x  2 root root      4096 Mar 31  2024/sbin.usr-is-merged
drwxr-xr-x 16 root root      4096 Apr 24 16:17 snap
drwxr-xr-x  2 root root      4096 Aug  5  2025 srv
-rw-----  1 root root 4294967296 Apr 21 20:29 swap.img
dr-xr-xr-x 13 root root         0 Sep  1 17:44 sys
drwxrwxrwt 20 root root      4096 Sep  1 19:45 tmp
drwxr-xr-x 12 root root      4096 Aug  5  2025 usr
drwxr-xr-x 14 root root      4096 Apr 21 20:31 var
```

Real-world use:

Security investigators inspect / to understand the overall filesystem structure and identify unusual directories or mount points.

Exercise 4 – Linux Filesystem Hierarchy Investigation

Investigate the following locations:

/home
/root
/etc
/var
/var/log
/tmp
/usr
/boot
/dev
/proc

For each location:

1. Examine its contents.
2. Identify at least three entries, where available.
3. Determine the likely purpose of the directory.

Directory	Example contents	Purpose
/home	claudio/, ubuntu/	Personal data and settings for each non-root user account.
/root	.bashrc, .config/, .profile, .ssh/	Home directory of the root (superuser) account, kept separate from /home.
/etc	.java/, .pwd.lock, ImageMagick-6/, LatexMk, PackageKit/	System-wide configuration files for the OS, network, users, and services.
/var	backups/, cache/, lib/, local/, lock -> /run/lock	Variable data that grows or changes while the system runs.

/var/log	README, alternatives.log, apt/, bootstrap.log, btmp	Log files for the system and applications.
/tmp	hsperfdata_root/, node-compile-cache/, phantomjs/	Temporary files, world-writable, cleared on reboot.
/usr	bin/, etc/, games/, include/, lib/	Bulk of installed user-space programs, libraries, and docs.
/boot	(empty in this container)	Kernel image and bootloader config on a normal machine.
/dev	autofs, console, cpu/, cpu_dma_latency, fd -> /proc/self/fd	Device files representing hardware and virtual devices.
/proc	numbered PID dirs, cpuinfo, meminfo	Virtual filesystem exposing live kernel and process info.

Exercise 5 – Find Your Home Directory

Without assuming its location:

1. Determine your username.

```
msis@master:/$ whoami
msis
```

2. Determine your home directory.

```
msis@master:/$ echo "$HOME"
/home/msis
```

3. Navigate to your home directory.

```
msis@master:/$ cd home
msis@master:/home$
```

4. Display its absolute path.

```
msis@master:/home$ pwd
/home
```

5. List its normal contents.

```
msis@master:/home$ ls
administrator  alice  bob  charlie  clusteruser  msis
```

6. List all contents, including hidden files.

```
msis@master:/home$ ls -la
total 32
drwxr-xr-x  8 root          root          4096 Sep  1 10:07 .
drwxr-xr-x 24 root          root          4096 Aug 11 13:50 ..
drwxr-x---  6 administrator administrator 4096 Jun 19 15:28 administrator
drwxr-x---  2 alice          alice          4096 Sep  1 10:07 alice
drwxr-x---  2 bob            bob            4096 Sep  1 10:07 bob
drwxr-x---  2 charlie        charlie        4096 Sep  1 10:07 charlie
drwxr-x---  6 clusteruser    clusteruser    4096 Aug  7 12:42 clusteruser
drwxr-x--- 36 msis           msis           4096 Sep  1 09:37 msis
```

Real-world use:

Security analysts may examine a user's home directory for scripts, SSH configuration, shell history, downloaded files, or persistence mechanisms.

Exercise 6 – Hidden Files Investigation

Examine your home directory.

Compare:

`ls`

`ls -a`

Identify at least 2 hidden files/directories.

```

msis@master:/home$ ls
administrator  alice  bob  charlie  clusteruser  msis
msis@master:/home$ ls -la
total 32
drwxr-xr-x  8 root          root          4096 Sep  1 10:07 .
drwxr-xr-x 24 root          root          4096 Aug 11 13:50 ..
drwxr-x---  6 administrator administrator 4096 Jun 19 15:28 administrator
drwxr-x---  2 alice         alice         4096 Sep  1 10:07 alice
drwxr-x---  2 bob          bob          4096 Sep  1 10:07 bob
drwxr-x---  2 charlie      charlie      4096 Sep  1 10:07 charlie
drwxr-x---  6 clusteruser  clusteruser  4096 Aug  7 12:42 clusteruser
drwxr-x--- 36 msis         msis         4096 Sep  1 09:37 msis

```

1. How does Linux identify a hidden file?

A filename beginning with `.` is normally treated as hidden

2. Why does a normal `ls` not display these files?

`ls` hides dot-prefixed entries by default.

3. What do ``.`` and ``.`` mean?

`.` —> current directory

`..` —> parent directory

4. Can a hidden file still be accessed if its name is known?

Yes

5. From a cybersecurity perspective, why should hidden files not automatically be considered safe or malicious?

Hidden does not mean safe.

For example, an attacker might place malicious scripts or persistence mechanisms in hidden locations such as:

`~/.config/`

`~/.ssh/`

`~/.local/`

Real-world scenario:

An incident responder notices an unexpected .config subdirectory containing an unfamiliar executable. They investigate its timestamps, permissions, contents, and whether it is executed automatically.

Exercise 7 – Build a Small Filesystem Hierarchy

Explore: mkdir

Hint: you can create this hierarchy in a single command

Create the following files:

cyber_lab/evidence/network/traffic.txt
cyber_lab/evidence/system/processes.txt
cyber_lab/logs/security.log
cyber_lab/scripts/check.sh
cyber_lab/reports/investigation.txt

```
mkdir -p cyber_lab/evidence/network \  
    cyber_lab/evidence/system \  
    cyber_lab/logs \  
    cyber_lab/scripts \  
    cyber_lab/reports
```

Use appropriate Linux commands to create the structure.

Verify that the required hierarchy has been created successfully.

```
msis@master:~$ tree cyber_lab  
locales-launch: Data of en_US locale not found, generating, please wait...  
cyber_lab  
├── evidence  
│   ├── network  
│   │   └── traffic.txt  
│   └── system  
│       └── processes.txt  
├── logs  
│   └── security.log  
├── reports  
│   └── investigation.txt  
├── scripts  
│   └── check.sh  
└──  
12 directories, 0 files
```

Real-world use:

Incident responders organize evidence by category—network evidence, system evidence, logs, scripts, and reports.

Exercise 8 – Absolute and Relative Paths

Assume your current directory is:

`~/cyber_lab`

Perform the following:

- 1. Navigate to ``evidence/network`` using a relative path.**
- 2. Navigate to ``/etc`` using an absolute path.**
- 3. Return to your home directory using the shortest possible command.**
- 4. Navigate to ``cyber_lab/reports``.**
- 5. Move one directory upward.**
- 6. Display the current directory after every operation.**

```
msis@master:~$ cd cyber_lab
msis@master:~/cyber_lab$ pwd
/home/msis/cyber_lab
msis@master:~/cyber_lab$ #1
msis@master:~/cyber_lab$ cd evidence/network
msis@master:~/cyber_lab/evidence/network$ pwd
/home/msis/cyber_lab/evidence/network
msis@master:~/cyber_lab/evidence/network$ #2
msis@master:~/cyber_lab/evidence/network$ cd /etc
msis@master:/etc$ pwd
/etc
msis@master:/etc$ #3
msis@master:/etc$ cd ~
msis@master:~$ #4
msis@master:~$ cd ~/cyber_lab/reports
msis@master:~/cyber_lab/reports$ pwd
/home/msis/cyber_lab/reports
msis@master:~/cyber_lab/reports$ #5
msis@master:~/cyber_lab/reports$ cd ..
msis@master:~/cyber_lab$ pwd
/home/msis/cyber_lab
```

\

* What is an absolute path?

Absolute path: It's the full address of a file or folder, starting all the way from /. No matter where you currently are in the terminal, it always points to the same place — like /etc/passwd.

* What is a relative path?

Relative path: It's a shortcut path based on where you're standing right now. It only makes sense from your current folder — so reports means different things depending on where you type it from

* What is the difference between `/` and `~`?

Difference between / and ~: / is the very top of the whole filesystem — everything on the system lives under it. ~ is just a quick way to say "my home folder" (like /root or /home/alice) — it's personal to whoever's logged in, not a fixed location like /

Exercise 9 – Identify File Types

Investigate the following:

`/etc/passwd`

`/etc`

`/bin/bash`

`/dev/null`

`/proc`

Also investigate the files created in Exercise 7.

`file <filename>`

`ls -l <filename>`

Object	Type
<code>/etc/passwd</code>	Regular file (plain text)
<code>/etc</code>	Directory
<code>/bin/bash</code>	Regular file (ELF executable binary)
<code>/dev/null</code>	Character special (device) file
<code>/proc</code>	Directory (virtual, not on disk)
<code>security.log</code>	Regular file (empty)
<code>check.sh</code>	Regular file (empty)

```
msis@master:~$  
file /etc/passwd  
file /etc  
file /bin/bash  
file /dev/null  
file /proc  
/etc/passwd: ASCII text  
/etc: directory  
/bin/bash: ELF 64-bit LSB pie executable, x86-64, version 1 (SYSV), dynamically linked, interpreter /lib64/ld-linux-x86-64.so.2, BuildID[sha1]=2f77b36371c214e11678c7d9d92727e9a49f626b,  
for GNU/Linux 3.2.0, stripped  
/dev/null: character special (1/3)  
/proc: directory  
msis@master:~$ ls -l /etc/passwd  
ls -ld /etc  
ls -l /bin/bash  
ls -l /dev/null  
ls -ld /proc  
-rw-r--r-- 1 root root 3577 Sep  1 14:13 /etc/passwd  
drwxr-xr-x 155 root root 12288 Sep  1 18:03 /etc  
-rwxr-xr-x 1 root root 1446024 Mar 31  2024 /bin/bash  
crw-rw-rw- 1 root root 1, 3 Sep  1 17:44 /dev/null  
dr-xr-xr-x 546 root root 0 Sep  1 17:44 /proc
```

Exercise 10 – Examine File Metadata

Use:

`~/cyber_lab/reports/investigation.txt`

Add a small amount of text to the file.

Examine it using:

`ls -l`
`stat investigation.txt`

Identify:

- * Filename
- * File size
- * File type
- * Owner
- * Group
- * Inode number
- * Access time
- * Modification time
- * Change time

```
msis@master:~/cyber_lab/reports$ ls -l investigation.txt
-rw-rw-r-- 1 msis msis 28 Sep  2 13:37 investigation.txt

msis@master:~/cyber_lab/reports$ stat investigation.txt
  File: investigation.txt
  Size: 28             Blocks: 8          IO Block: 4096   regular file
Device: 259,7   Inode: 12192134    Links: 1
Access: (0664/-rw-rw-r--)  Uid: ( 1000/   msis)   Gid: ( 1000/   msis)
Access: 2026-09-02 13:37:55.915790335 +0530
Modify: 2026-09-02 13:37:49.817579321 +0530
Change: 2026-09-02 13:37:49.817579321 +0530
 Birth: 2026-09-02 13:37:49.816697274 +0530
```

1. What is metadata?

Metadata is information about a file, such as its size, owner, permissions, and timestamps.

2. Is metadata the same as file content?

No. Metadata describes the file, while content is the actual data inside the file.

3. Which timestamp changes when file contents are modified?

The Modification Time (mtime) changes when the file content is modified.

Real-World Use:

This exercise is useful in system administration and cybersecurity investigations. Commands such as `ls -l`, `file`, and `stat` help identify a file's type, owner, permissions, size, inode number, and timestamps. Investigators can use this information to find suspicious files and determine when a file was accessed or modified.

Exercise 11 – Explore `/etc`

Investigate:

`/etc`

...

Locate:

`passwd`

`group`

`hostname`

`Hosts`

```
msis@master:~/cyber_lab/reports$ ls -l /etc/passwd
-rw-r--r-- 1 root root 3577 Sep  1 14:13 /etc/passwd
msis@master:~/cyber_lab/reports$ ls -l /etc/group
-rw-r--r-- 1 root root 1433 Sep  1 18:03 /etc/group
msis@master:~/cyber_lab/reports$ ls -l /etc/hostname
-rw-r--r-- 1 root root 7 Aug  7 11:24 /etc/hostname
msis@master:~/cyber_lab/reports$ ls -l /etc/hosts
-rw-r--r-- 1 root root 606 Aug  7 11:32 /etc/hosts
```

Note: Without modifying any system files, examine their contents where permitted.

1. What kind of information is primarily stored under `/etc`?

`/etc` mainly stores system-wide configuration files.

2. What information does `/etc/passwd` contain?

It contains user account information such as usernames, user IDs, home directories, and login shells.

3. What information does `/etc/hostname` contain?

It contains the name (hostname) of the Linux system.

4. Why is `/etc` security-sensitive?

It is security-sensitive because it contains important system and security configuration files.

Exercise 12 – Explore Linux Log Storage

Investigate:

`/var/log`

``

Identify:

*** Available log files**

Command: `find /var/log -maxdepth 1 -type d`

*** Available log directories**

Command: `find /var/log -maxdepth 1 -type d`

*** Recently modified entries**

Command: `ls -lt /var/log | head`

*** Entries you can read**

Command: `find /var/log -type f -readable`

*** Entries you cannot read as a normal user**

Command: `find /var/log -type f ! -readable`

Answer:

1. Why does Linux maintain logs?

Linux maintains logs to record system, application, and security events.

2. Why are logs useful during incident investigation?

Logs help investigators determine what happened, when it happened, and who or what was involved.

3. Why might access to some logs be restricted?

Some logs contain sensitive security information, so access is restricted to prevent unauthorized access or modification.

Exercise 13 – Investigate `/tmp`

Explore `/tmp` without modifying files belonging to other users.

Determine:

1. Whether files/directories currently exist.

```
find /tmp -maxdepth 1 -type f
```

```
find /tmp -maxdepth 1 -type d
```

2. Whether hidden files exist.

```
ls -la /tmp
```

or

```
msis@master:~/cyber_lab/reports$ find /tmp -maxdepth 1 -name ".*"
/tmp/.X11-unix
/tmp/.X1025-lock
/tmp/.ICE-unix
/tmp/.X1024-lock
/tmp/.font-unix
/tmp/.XIM-unix
/tmp/.X0-lock
/tmp/.X1-lock
```

3. Who owns some of the entries.

```
ls -lah /tmp
```

```
-rw-r--r--  1 jenkins jenkins 4.7M Sep  2 13:10 winstone14342431687842909196.jar
-r--r--r--  1 msis    msis      11 Sep  2 13:10 .X0-lock
-r--r--r--  1 gdm     gdm      11 Sep  2 13:10 .X1024-lock
-r--r--r--  1 gdm     gdm      11 Sep  2 13:10 .X1025-lock
drwxrwxrwt  2 root    root      4.0K Sep  2 13:10 .X11-unix
-r--r--r--  1 msis    msis      11 Sep  2 13:10 .X1-lock
drwxrwxrwt  2 root    root      4.0K Sep  2 13:10 .XIM-unix
```

4. Which entries were modified recently.

```
msis@master:~/cyber_lab/reports$ ls -lat /tmp | head
total 4892
drwxrwxrwt 20 root    root    4096 Sep  2 13:44 .
drwx----- 3 root    root    4096 Sep  2 13:44 systemd-private-d0c941af35fa4
3ea888309d8c88f5abc-fwupd.service-tGmesg
drwx----- 4 root    root    4096 Sep  2 13:11 snap-private-tmp
drwxrwxrwt 2 root    root    4096 Sep  2 13:10 .X11-unix
-r--r--r-- 1 msis    msis     11 Sep  2 13:10 .X1-lock
-r--r--r-- 1 msis    msis     11 Sep  2 13:10 .X0-lock
drwxrwxrwt 2 root    root    4096 Sep  2 13:10 .ICE-unix
drwx----- 3 root    root    4096 Sep  2 13:10 systemd-private-d0c941af35fa4
3ea888309d8c88f5abc-colord.service-99VIOw
-r--r--r-- 1 gdm     gdm      11 Sep  2 13:10 .X1025-lock
```

> Why can temporary directories be relevant during a cybersecurity investigation?

Temporary directories can contain suspicious scripts, downloaded files, malware, or temporary data created during an attack.

Real-World Use

Cybersecurity investigators examine /tmp because attackers or malicious programs may store temporary scripts and files there. Checking file names, owners, and modification times can help identify suspicious activity and build an investigation timeline.

Exercise 14 – Explore the `/proc` Filesystem

Investigate:

```
/proc
'''
```

Examine:

```
cat /proc/cpuinfo
cat /proc/meminfo
'''
```

/proc/cpuinfo provides information about the system's CPU, such as the processor model, number of processors/cores, architecture, and CPU features.
/proc/meminfo provides information about system memory, including total memory, available memory, free memory, and swap information.

Then identify your current shell PID:

echo \$\$

'''

```
msis@master:~/cyber_lab/reports$ echo $$  
8592
```

Check whether a corresponding directory exists under `/proc`.

Information about current shell process

```
msis@master:~/cyber_lab/reports$ ls -ld /proc/$$  
dr-xr-xr-x 9 msis msis 0 Sep  2 13:25 /proc/8592
```

1. What information does `/proc/cpuinfo` provide?

It provides information about the system's processors and CPU features.

2. What information does `/proc/meminfo` provide?

It provides information about system memory and memory usage.

3. Why do directories with numbers appear inside `/proc`?

Numbered directories represent running processes, and each number is a Process ID (PID).

4. Is `/proc` an ordinary disk directory?

No. /proc is a virtual filesystem created by the Linux kernel to provide system and process information.

5. Why could `/proc` be useful during security investigation?

/proc can provide information about running processes, system activity, memory, and process details that may help identify suspicious activity.

Real-World Use :

Security analysts use /proc to investigate running processes and system information. For example, if a suspicious process is found, an investigator can examine its PID directory under /proc to gather information about the process.

Exercise 15 – Explore `/dev`

/dev

...

Examine:

`ls -l /dev/null`

`ls -l /dev/zero`

...

```
msis@master:~/cyber_lab/reports$ ls -l /dev/null
crw-rw-rw- 1 root root 1, 3 Sep  2 13:10 /dev/null
msis@master:~/cyber_lab/reports$ ls -l /dev/zero
crw-rw-rw- 1 root root 1, 5 Sep  2 13:10 /dev/zero
```

1. What does `/dev` represent?

/dev contains special files that provide access to devices and hardware in Linux.

2. Are the entries ordinary user files?

No. Many entries in /dev are special device files rather than ordinary user files

3. Determine the file type of `/dev/null`.

/dev/null is a character special device

4. Why does Linux expose devices through the filesystem?

Linux exposes devices through the filesystem so programs can interact with hardware and system devices using standard file operations.

```
msis@master:~/cyber_lab/reports$ #1
msis@master:~/cyber_lab/reports$ ls -la /dev | head -5
total 4
drwxr-xr-x 20 root root          4980 Sep  2 13:10 .
drwxr-xr-x 24 root root          4096 Aug 11 13:50 ..
crw----- 1 root root           10, 262 Sep  2 13:10 acpi_thermal_rel
crw-r--r-- 1 root root           10, 235 Sep  2 13:10 autofs
msis@master:~/cyber_lab/reports$ #2
msis@master:~/cyber_lab/reports$ ls -l /dev/null /dev/zero
crw-rw-rw- 1 root root 1, 3 Sep  2 13:10 /dev/null
crw-rw-rw- 1 root root 1, 5 Sep  2 13:10 /dev/zero
msis@master:~/cyber_lab/reports$ #3
msis@master:~/cyber_lab/reports$ file /dev/null
/dev/null: character special (1/3)
msis@master:~/cyber_lab/reports$ #4
msis@master:~/cyber_lab/reports$ ls -l /dev | head -5
total 0
crw----- 1 root root           10, 262 Sep  2 13:10 acpi_thermal_rel
crw-r--r-- 1 root root           10, 235 Sep  2 13:10 autofs
drwxr-xr-x 2 root root           720 Sep  2 13:10 block
crw----- 1 root root           10, 234 Sep  2 13:10 btrfs-control
```

Real-World Use

/dev is important in system administration and cybersecurity investigations because it shows the devices and special interfaces available to the system. Investigators can examine unusual device files or determine whether a suspicious process is interacting with system devices.

Exercise 16 – Filesystem Space Investigation

Determine:

1. Total filesystem capacity.
2. Used space.
3. Available space.
4. Percentage of filesystem utilization.

```
msis@master:~/cyber_lab/reports$ #1#2#3#4
msis@master:~/cyber_lab/reports$ df -h
Filesystem      Size  Used Avail Use% Mounted on
tmpfs            1.6G  2.8M  1.6G   1% /run
/dev/nvme0n1p7  233G   64G  158G  29% /
tmpfs            7.7G  4.0K  7.7G   1% /dev/shm
tmpfs            5.0M  8.0K  5.0M   1% /run/lock
efivarfs         246K  180K   62K  75% /sys/firmware/efi/efivars
tmpfs            7.7G   0  7.7G   0% /run/qemu
/dev/nvme0n1p1  196M   77M  120M  40% /boot/efi
tmpfs            1.6G  2.5M  1.6G   1% /run/user/1000
```

5. Space occupied by your home directory.
6. Space occupied by the `cyber\_lab` directory.

```
msis@master:~$ #5
msis@master:~$ du -sh ~
45G    /home/msis
msis@master:~$ #6
msis@master:~$ du -sh cyber_lab
48K    cyber_lab
```

Use appropriate combinations of:

df

df -h

du
du -h
du -sh

> Explain the difference between `df` and `du`.

df tells you how much space the filesystem has and how much is free; **du** tells you where that space is being used.

Exercise 17 – Cybersecurity Investigation Scenario

Assume that you are investigating a Linux machine after suspicious activity has been reported.

You are initially asked to examine only the filesystem.

Identify which of the following locations you would investigate and explain **why****:**

/home
/etc
/var/log
/tmp
/proc
/dev
/root

Submit the following:

- 1. Commands used for each exercise.**
- 2. Relevant output or screenshots.**
- 3. Observation tables.**
- 4. Disk → Partition → Filesystem → Mount Point diagram.**
- 5. Answers to analysis questions.**

Location	Investigate ?	Why	Security-relevant info expected
/home	Yes	Direct evidence of user activity	.bash_history, SSH keys, downloaded tools, browser artifacts
/etc	Yes	Defines what the system is set up to allow	Unauthorized accounts, sudoers rules, cron jobs, SSH config changes
/var/log	Yes	Primary timeline of what already happened	Login history (wtmp/btmp), failed logins, auth logs, install logs
/tmp	Yes	World-writable, wiped on reboot — malware staging spot	Unexpected scripts/binaries, hidden files, mismatched ownership
/proc	Yes	Live view of what's happening right now	Real cmdline, actual running binary (exe), open network sockets
/dev	Sometimes	Normally routine, but can be abused	Unexpected device nodes, data disguised as a device file
/root	If escalation suspected	Shows what an attacker did with elevated access	Root's shell history, SSH keys, config changes as superuser